IoT Cat Feeder

An automatic, portion-controlled cat feeder with a web dashboard on the ESP8266

Software Requirements Specification

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Chapter 1 Introduction

1.1 Purpose

This document specifies the requirements for the IoT Cat Feeder, version 1.0.0. It is written for the project advisor as the acceptance basis, for the developer as the input to the design, and for the examiner as the evidence that each requirement has a stated, checkable condition.

The document follows the structure of ISO/IEC/IEEE 29148 for software requirements. The section order is adapted to the size of the project: the standard separates product functions from detailed requirements, which this document keeps, and it separates verification from the requirement text, which this document keeps as a matrix in Section 2.10.

1.2 Scope

The product is a single-board appliance that dispenses dry cat food on a schedule and on demand, and reports its state through a web page served by the same board.

Table 1. System identification

Item	Value
Product name	IoT Cat Feeder
Version specified	1.0.0
Platform	NodeMCU v3, ESP8266EX, 80 MHz, 4 MB flash
Runtime	Arduino core for ESP8266, single-threaded main loop
Network role	Wi-Fi station client with an embedded HTTP server on port 80
Users	One owner, one cat
Operating environment	Indoor, 2.4 GHz Wi-Fi, 5 V 2 A supply

The product does not store data outside the device, does not authenticate users, and does not depend on any external service other than a network time server.

1.3 Product Perspective

The device is a standalone system. It replaces the manual act of pouring food, not a software component inside another product.

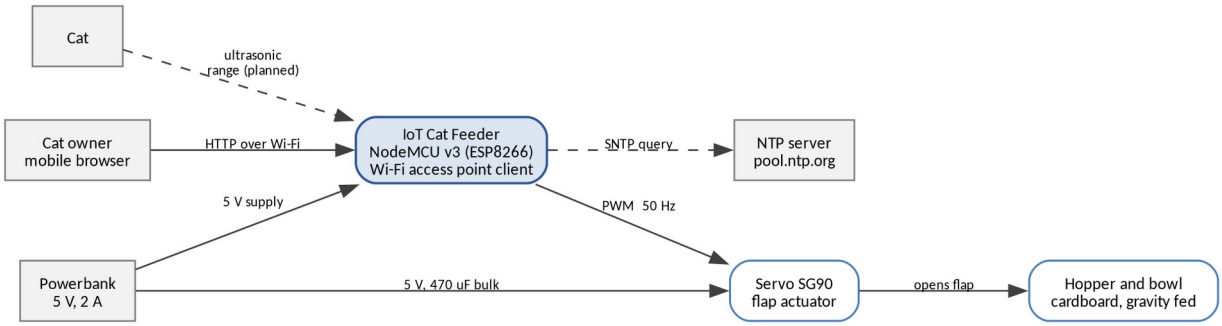


Figure 1. System context. Dashed elements are planned.

The owner is the only human actor. The clock is an external service that provides the time reference. The cat is an environmental actor whose position may become an input once the presence requirement is delivered.

1.4 Product Functions

Table 2. Product functions

ID	Function	Summary	Status
F1	Scheduled feeding	Dispenses one meal at each configured time, using network time.	Delivered
F2	On-demand feeding	Dispenses one meal when the owner presses the dashboard button.	Delivered
F3	Status and history	Reports flap state, daily count, last meal, and link quality.	Delivered
F4	Presence-triggered feeding	Dispenses a meal when a cat is detected near the bowl.	Target
F5	Portion and level control	Dispenses a measured mass and reports hopper level.	Target

1.5 User Characteristics

Table 3. User classes

User class	Technical level	Characteristics and needs
Owner	Non-technical	Uses a phone browser. Expects a single page with a single action. Cannot be asked to install software, create an account, or read a manual.
Maintainer	Technical	Rebuilds and reflashes the device. Needs the wiring, the pin map, the library list, and the credentials file documented.
Cat	None	Interacts physically. Must not be able to open the flap or be caught by it.

1.6 Limitations

- History is held in RAM. A reboot clears the daily count.
- Any device on the same Wi-Fi network can issue a feeding command, because there is no authentication.
- Only dry food with a free-flowing particle size can be dispensed.
- The device cannot detect that the bowl is already full before dispensing.
- Only one feeder and one hopper are supported.
- The system does not operate without a 5 V supply.

1.7 Assumptions and Dependencies

Table 4. Assumptions and dependencies

ID	Statement	Effect if false
A-01	The Wi-Fi network is 2.4 GHz and reachable from the device position.	The device cannot join. The dashboard is unreachable.
A-02	A network time server is reachable at least once after each boot.	Scheduled feeding is disabled. Manual feeding still works.
A-03	The hopper holds at least 300 g of dry food.	Meals become short. The level requirement reports the shortage.
A-04	The servo can lift the flap without stalling.	Brownout or a failed meal. R-01 in the proposal covers the mitigation.
A-05	The owner's phone and the device are on the same subnet.	The dashboard cannot be reached by name or address.

ID	Statement	Effect if false
A-06	The load cell can be mounted on a rigid platform.	Portion control is descoped and recorded as future work.

1.8 Acronyms and Definitions

1.8.1 Acronyms

Table 5. Acronyms

Acronym	Meaning
AP	Access point
GPIO	General purpose input and output
HTTP	Hypertext Transfer Protocol
HX711	24-bit load cell amplifier
JSON	JavaScript Object Notation
NTP, SNTP	Network Time Protocol and its simplified form
PWM	Pulse width modulation
SRS	Software Requirements Specification
SDD	Software Design Description
UART	Universal asynchronous receiver and transmitter

1.8.2 Definitions

Table 6. Definitions

Term	Definition
Meal	One dispense cycle, defined as one flap open event followed by an automatic close.
Flap	The servo-driven cover that gates the hopper outlet.
Hopper	The container that holds dry food above the flap.
Schedule minute	A minute of local time that appears in the configured meal table.
Source	The trigger of a meal: `manual`, `schedule`, or `presence`.
Minute key	An integer that identifies one calendar minute, used to prevent duplicate meals.
Target feature	A requirement that is specified and designed but ships only if the parts pass test before 2 November 2026.

1.9 References

Table 7. References

Reference	Use in this document
ISO/IEC/IEEE 29148:2018, Requirements engineering	Structure of the requirement sections.
IEEE Std 1016-2009, Software design descriptions	Structure of the companion SDD.
Arduino core for ESP8266, reference documentation	Behaviour of `ESP8266WebServer`, `configTime`, and `WiFi`.
HC-SR04 ultrasonic sensor datasheet	Range, beam angle, and supply voltage limits.
HX711 datasheet	Clock timing and gain selection.

Reference	Use in this document
SG90 servo datasheet	Stall current and torque limits.
Project rubric, Idea 50 / Circuit 30 / Doc 20	Coverage check in Appendix A.

1.10 Document Control

Table 8. Revision history

Version	Date	Author	Change
0.9	30 October 2026	C. Jaiman	First draft in Thai, requirements only.
1.0.0	4 November 2026	C. Jaiman	Baselined. Sensor requirements marked as target.

Chapter 2 Specification Requirements

2.1 Feature Description

2.1.1 F1: Scheduled Feeding

At each configured meal time the device dispenses one meal without any user action. The time reference comes from a network time server, so the schedule stays correct after a power cut without a battery-backed clock.

Table 9. F1 parameters

Parameter	Value
Meal times	08:00, 13:00, 18:00
Time zone	UTC+7, no daylight saving
Time sources	pool.ntp.org, time.nist.gov
Duplicate suppression	One meal per calendar minute
Behaviour before time is valid	Scheduler disabled, dashboard shows --:--:--

2.1.2 F2: On-Demand Feeding

The owner presses a button on the dashboard and receives a meal within 1.5 seconds. The request is a POST to /feed. The button disables itself for 1.3 seconds after the press so that a double tap does not produce two overlapping cycles.

Table 10. F2 parameters

Parameter	Value
Endpoint	POST /feed
Response	200 with {"ok":true}
Flap open duration	900 ms
Client lockout	1300 ms
Accepted methods	POST only. Others return 404.

2.1.3 F3: Status and History

The dashboard shows five values and refreshes without user action. The values answer one question: was the cat fed today, and by what.

Table 11. F3 parameters

Displayed value	Source field	Format
Flap state	open	Open, or Closed
Meals today	count	Integer with unit
Last meal	last, source	HH:MM with trigger in brackets
Device time	time	HH:MM:SS
Configured schedule	schedule	Three times separated by a middle dot
Link quality	rssi, ip	dBm and dotted quad

2.1.4 F4: Presence-Triggered Feeding (target)

An ultrasonic sensor aimed at the approach path detects a cat within 60 cm. A confirmed detection dispenses one meal, subject to a cooldown and a daily cap. Confirmation requires three consecutive readings inside range, which prevents a single stray echo from feeding the cat.

Table 12. F4 parameters

Parameter	Value
Detection range	60 cm
Confirmation	3 consecutive readings
Cooldown after a presence meal	30 minutes
Daily cap	Configurable, default 6
Sensor	HC-SR04 or HC-SR04P, echo divided to 3.3 V

2.1.5 F5: Portion and Level Control (target)

A load cell under the bowl measures the mass dispensed. The controller closes the flap when the target mass is reached, or after a timeout, whichever comes first. A second ultrasonic sensor aimed down into the hopper estimates the remaining food.

Table 13. F5 parameters

Parameter	Value
Target portion	20 g per meal, adjustable
Tolerance	plus or minus 15 percent
Flap timeout	3 s, then close regardless of mass
Level reporting	Percentage of usable hopper depth
Amplifier gain	Channel A, gain 128

2.2 Use Case Diagram

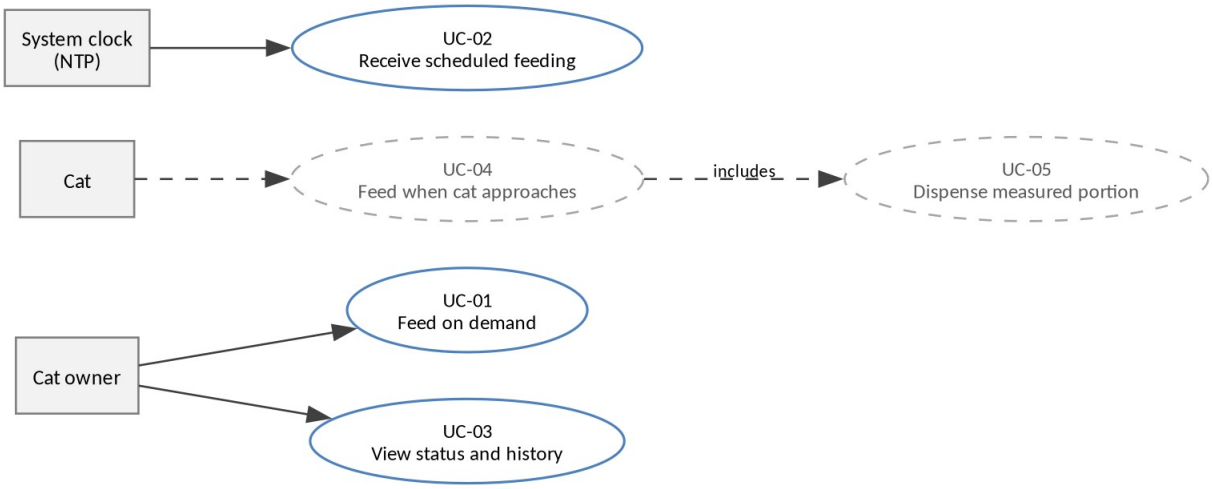


Figure 2. Use cases. Dashed use cases are targets and are not part of the delivered baseline.

2.3 Use Case Description

2.3.1 UC-01: Feed the Cat on Demand

Table 14. UC-01: Feed the cat on demand

Item	Description
Actor	Owner
Goal	Dispense one meal immediately.
Precondition	The device is powered and reachable on the network. A browser has loaded the dashboard.
Trigger	The owner presses the feeding button.
Main flow	1. The browser sends POST /feed. 2. The device validates the method. 3. The device records the meal with source manual. 4. The device drives the servo to the open angle. 5. The device returns {"ok":true}. 6. After 900 ms the main loop closes the flap.
Alternate flow	A1. The request times out: the browser shows a connection failure, and the flap still closes because the close is driven locally.
Postcondition	One meal is counted. The flap is closed.
Frequency	Up to several times per day.

2.3.2 UC-02: Receive a Scheduled Feeding

Table 15. UC-02: Receive a scheduled feeding

Item	Description
Actor	System clock, provided by NTP
Goal	Dispense one meal at a configured time without user action.
Precondition	The device has a valid time, meaning the epoch is later than 9 September 2020.
Trigger	Local time reaches a minute in the meal table.
Main flow	1. The main loop reads the local time. 2. The minute key is computed. 3. If the key equals the previous key, the loop exits. 4. If the minute matches a meal time, one meal is dispensed with source schedule. 5. The flap closes after 900 ms.
Alternate flow	A1. Time is not yet valid: the loop exits without dispensing. A2. The clock jumps backward after a resync: a repeated minute key still produces at most one meal.
Postcondition	One meal is counted with source schedule.
Frequency	Three times per day.

2.3.3 UC-03: View Status and History

Table 16. UC-03: View status and history

Item	Description
Actor	Owner
Goal	Confirm that the cat has been fed and inspect the device clock.

Item	Description
Precondition	The device is reachable.
Trigger	The owner opens the dashboard, or the two-second refresh fires.
Main flow	1. The browser sends GET /status. 2. The device serialises eight fields. 3. The page updates five display rows and the link footer.
Alternate flow	A1. The request fails: the footer reports a lost connection and the previous values stay on screen.
Postcondition	No state changes on the device.
Frequency	Continuous while the page is open.

2.3.4 UC-04: Feed When the Cat Approaches (target)

Table 17. UC-04: Feed when the cat approaches

Item	Description
Actor	Cat
Goal	Dispense a meal when the cat is at the bowl and the cooldown has expired.
Precondition	The presence sensor is connected and calibrated. The daily cap is not reached.
Trigger	Three consecutive distance readings below 60 cm.
Main flow	1. The loop polls the sensor. 2. The median distance is computed. 3. The confirmation counter increments. 4. A meal is dispensed with source presence. 5. The cooldown starts.
Alternate flow	A1. The cat leaves before confirmation: the counter resets. A2. Cooldown is active: the request is ignored and no meal is counted.
Postcondition	One meal is counted. The presence cooldown is running.
Frequency	At most once per 30 minutes.

2.3.5 UC-05: Dispense a Measured Portion (target)

Table 18. UC-05: Dispense a measured portion

Item	Description
Actor	System
Goal	Deliver a mass of food within tolerance of the target.
Precondition	The load cell is calibrated at zero with an empty bowl.
Trigger	Any feeding path starts.
Main flow	1. The flap opens. 2. The amplifier is sampled ten times per second. 3. The mass delta is compared with the target. 4. The flap closes when the target is reached.
Alternate flow	A1. The timeout of 3 s expires: the flap closes and the meal is recorded with a shortfall flag.
Postcondition	The dispensed mass and the shortfall are recorded.
Frequency	Once per meal.

2.3.6 UC-06: Recover From a Network Loss

Table 19. UC-06: Recover from a network loss

Item	Description
Actor	System
Goal	Return to the network without a reboot.
Precondition	The device was connected and lost the link.
Trigger	The connection state changes to disconnected.
Main flow	1. The loop checks the connection state. 2. A retry is issued no more than once every 5 seconds. 3. On success the local address is printed once to the serial console.
Alternate flow	A1. The network is absent: retries continue indefinitely at the same rate.
Postcondition	The dashboard is reachable again. No meal state is lost.
Frequency	On every disconnection.

2.4 Activity Diagram

Figure 3 shows one iteration of the main loop. The close of the flap is evaluated on every iteration and does not depend on an HTTP request completing.

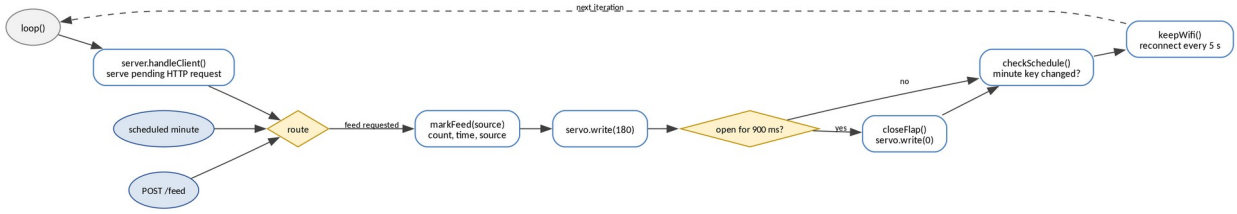


Figure 3. Main loop activity.

2.5 Functional Requirements

Priority uses MoSCoW. Verification uses one of T for test, D for demonstration, A for analysis, or I for inspection.

Table 20. Functional requirements

ID	Requirement	Priority	Verification	Status
FR-01	The device shall dispense one meal at 08:00, 13:00, and 18:00 local time.	Must	T	Delivered
FR-02	The device shall obtain local time from a network time server and shall not evaluate the schedule until the time is valid.	Must	T	Delivered
FR-03	The device shall dispense at most one scheduled meal per calendar minute.	Must	T	Delivered
FR-04	The device shall dispense one meal when it receives POST /feed.	Must	T	Delivered
FR-05	The device shall close the flap within 1000 ms of opening it, on every code path.	Must	T	Delivered
FR-06	The device shall reject any method other than POST on /feed.	Must	T	Delivered
FR-07	The device shall return the eight status fields defined in Table 12 on GET /status.	Must	T	Delivered

ID	Requirement	Priority	Verification	Status
FR-08	The dashboard shall refresh its values every 2 seconds without user action.	Must	D	Delivered
FR-09	The device shall reset the daily meal count when the local date changes.	Must	T	Delivered
FR-10	The device shall record the trigger source of every meal as manual, schedule, or presence.	Must	T	Delivered
FR-11	The device shall serve the dashboard page on GET / with no external asset.	Must	T	Delivered
FR-12	The device shall return 404 for every undefined path.	Must	T	Delivered
FR-13	The device shall detect a cat within 60 cm of the bowl after three consecutive readings.	Should	T	Target
FR-14	The device shall not dispense more than one presence-triggered meal in 30 minutes.	Should	T	Target
FR-15	The device shall dispense a mass within 15 percent of the configured target.	Could	T	Target
FR-16	The device shall report hopper level as a percentage of usable depth.	Could	D	Target
FR-17	The device shall persist the daily history across a reboot.	Could	T	Future work
FR-18	The device shall refuse a meal when the configured daily cap is reached.	Should	T	Future work

2.6 External Interface Requirements

2.6.1 User Interface

One page, one column, maximum width 420 px, in Thai. The page uses a light background and a single accent colour. It has no navigation, no form, and no settings panel.

Table 21. Dashboard elements

Element	Type	Behaviour
Title block	Static text	Names the device and states the primary action.
Status card	Five text rows	Rebuilt from /status on every refresh.
Feed button	Button	Sends POST /feed, disables itself for 1300 ms.
Link footer	Text	Shows RSSI and IP, or a lost-connection message.

2.6.2 Software Interface

Table 22. HTTP interface

Method	Path	Request body	Response
GET	/	None	HTML, 200
GET	/status	None	JSON, 200, application/json; charset=utf-8
POST	/feed	None	JSON {"ok":true}, 200
Any	Other	None	Text, 404

Table 23. Status response fields

Field	Type	Range	Meaning
open	Boolean	true, false	Flap is open.
count	Integer	0 to 255	Meals counted since local midnight.
last	String	HH:MM or -	Local time of the last meal.
source	String	manual, schedule, presence, empty	Trigger of the last meal.
time	String	HH:MM:SS or --:--:--	Device local time.
schedule	String	Three times	Configured meal times.
ip	String	Dotted quad	Device address.
rssi	Integer	-90 to -30	Signal strength in dBm.

2.6.3 Hardware Interface

Table 24. Hardware interface

Device	Signal	Pin	Electrical notes	Status
Servo SG90	PWM	D5, GPIO14	50 Hz, 1 to 2 ms pulse. Powered from VIN, not 3V3.	Connected
Servo SG90	Supply	VIN, GND	5 V from powerbank. 470 uF across the rail.	Connected
HC-SR04	Trig	To be assigned	3.3 V drive is accepted.	Target
HC-SR04	Echo	To be assigned	5 V output requires a divider for 3.3 V logic.	Target
HX711	DT, SCK	To be assigned	Two digital lines. Avoid GPIO0, GPIO2, GPIO15 at boot.	Target
Status LED	Digital out	D4, GPIO2	Bring-up only. Active low on this board.	Test

2.7 Data Requirements

The device keeps one record in RAM. There is no database, no file system, and no remote store in the delivered baseline.

Table 25. Persistent and in-memory data

Name	Type	Lifetime	Purpose
Feeder state	Struct of six fields	Until reboot	Flap state, count, last meal, source, day.
Meal table	Constant array of three hours	Firmware constant	Schedule definition.
Minute key	Integer	Until reboot	Duplicate suppression for the scheduler.
Last connection attempt	Unsigned long	Until reboot	Rate limit for Wi-Fi retries.

Requirement FR-17 would add a flash record with one line per meal, written append-only, capped at the last 32 meals. That record is not part of version 1.0.0.

2.8 Quality Requirements

Table 26. Non-functional requirements

ID	Category	Requirement	Verification
NFR-PM-01	Performance	A dashboard request completes within 2 s on the same network.	T
NFR-PM-02	Performance	A manual meal starts within 1.5 s of the button press.	T
NFR-PM-03	Performance	Presence detection dispenses within 2 s of entering range.	T
NFR-RL-01	Reliability	The flap is never open longer than 3 s.	T
NFR-RL-02	Reliability	The device rejoins the network within 30 s of it returning, with no reboot.	T
NFR-RL-03	Reliability	Schedule timing drifts by no more than 1 minute over 24 hours.	T
NFR-RL-04	Reliability	A failed HTTP response does not leave the flap open.	A
NFR-UH-01	Usability	The dashboard is readable at 375 px width without zoom.	D
NFR-UH-02	Usability	A first-time user feeds the cat without instruction.	D
NFR-UH-03	Usability	The dashboard states the connection state when the link fails.	D
NFR-SEC-01	Security	Wi-Fi credentials are excluded from version control.	I
NFR-SEC-02	Security	Only documented methods and paths are served.	T
NFR-SEC-03	Security	The device exposes no administrative endpoint.	I
NFR-OP-01	Maintainability	The firmware builds from a documented library set.	I
NFR-OP-02	Maintainability	The pin map and wiring are documented in one page.	I
NFR-CP-01	Constraint	The device runs from a 5 V 2 A powerbank with no host computer.	T
NFR-CP-02	Constraint	The device operates on 2.4 GHz Wi-Fi only.	I
NFR-CP-03	Constraint	The firmware fits in the 4 MB flash of the NodeMCU v3.	I

2.9 Design Constraints

- The servo signal uses GPIO14 because the board exposes it as D5 with no boot-time conflict.
- GPIO16 is not usable for the load cell clock because it has no interrupt support on this core.
- The echo line of a 5 V ultrasonic sensor must be divided before it reaches a 3.3 V input.
- No blocking delay is permitted in the main loop, because the HTTP server is polled from the same loop.
- The dashboard must be complete in one response under 4 KB, because it is compiled into the binary.

2.10 Verification

Each requirement maps to one test case in the SDD. The matrix below states the method and the pass condition at requirement level.

Table 27. Verification matrix

Requirement group	Method	Pass condition
FR-01 to FR-03	Timed observation	Exactly three scheduled meals in 24 hours, each within 1 minute.
FR-04, FR-05	Manual test, 50 repetitions	Every repetition dispenses and closes. No flap stays open.
FR-06, FR-12	Request inspection	Non-POST and unknown paths return 404 with no side effect.
FR-07, FR-10, FR-11	Response inspection	All fields present, values match the device state.

Requirement group	Method	Pass condition
FR-08	Observation	The page changes without user action within 2 s.
FR-09	Clock rollover	The count returns to zero after local midnight.
FR-13, FR-14	Field test with the cat	16 of 20 detections correct. No second meal inside 30 minutes.
FR-15, FR-16	Scale comparison	Dispensed mass within 15 percent of target on 10 trials.
FR-17	Power cycle	History survives a reboot. Currently fails by design.
NFR-PM-01	Stopwatch	Median response under 2 s across 20 requests.
NFR-RL-02	Router power cycle	The device rejoins with no manual action.
NFR-SEC-01	Repository inspection	secrets.h is absent from the tree and from history.

Chapter A Rubric Coverage

The assessment rubric weights Idea 50, Circuit 30, and Doc 20. The table maps each criterion to the section that satisfies it.

Table 28. Rubric coverage

Criterion	Weight	Where it is answered
Concept and problem fit	Idea	Proposal 1.1 and 1.2
Innovation and impact	Idea	Proposal 2.1.1 and 2.1.3, SRS 2.1.5
UX and use-case clarity	Idea	Proposal 1.3, SRS 2.3 and 2.6.1
Architecture concept and feasibility	Idea	Proposal 2.2 and 3.3, SDD Chapter 2
Hardware implementation	Circuit	SRS 2.6.3, SDD 2.4
Software and firmware	Circuit	SRS 2.5, SDD 3.2
System integration and IoT connectivity	Circuit	SRS 2.6.2, SDD 3.5
SRS	Doc	This document
Proposal and abstract	Doc	Proposal Chapter 1
SDD	Doc	SDD Chapters 2 and 3
Output and discussion	Doc	Output chapter, to be written after testing
Photos and evidence	Doc	Test record, to be captured on 3 November
Conclusion and future work	Doc	Proposal 5.5, and the conclusion chapter